

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 05-Sep-2019

Revision Number 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Name Copper (Soluble)

Product No AC4029-AMP

Pure substance/mixture Mixture

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

Supplier

Fisher Scientific Korea	Thermo Fisher Scientific©
D5,D6, Incheon Airport Logistics	Water and Lab Products
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Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
 CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

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Skin Corrosion/Irritation
Serious Eye Damage/Eye Irritation
Specific target organ toxicity - (single exposure)

Category 2
Category 2
Category 3

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements



Signal Word

Warning

Hazard Statements

H315 - Causes skin irritation
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H336 - May cause drowsiness or dizziness

Precautionary Statements

Prevention

P261 - Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray
P264 - Wash face, hands and any exposed skin thoroughly after handling
P271 - Use only outdoors or in a well-ventilated area
P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection

Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P312 - Call a POISON CENTER or doctor/ physician if you feel unwell
P362 - Take off contaminated clothing and wash before reuse
P321 - Specific treatment (see supplemental first aid instructions on this label)
P332 + P313 - If skin irritation occurs: Get medical advice/ attention
P362 + P364 - Take off all contaminated clothing and wash it before reuse
P337 + P313 - If eye irritation persists: Get medical advice/ attention

Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
P405 - Store locked up

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

Repeated exposure may cause skin dryness or cracking

NFPA

Health
3

Flammability
0

Instability
0

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

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3.2. Mixtures

Component	Common Name	CAS-No	Index No.	Weight %
Water	No information available	7732-18-5	KE-35400	70 - 80
Ammonium Acetate	No information available	631-61-8	KE-01629	20 - 30
Sodium Bisulfite	No information available	7631-90-5	KE-31484	0.5 - 1
Ammonium Oxalate	No information available	1113-38-8	KE-13156	0.5 - 1
Acetone	No information available	67-64-1	KE-29367	0.5 - 1
Bathocuproine	No information available	52698-84-7	Not listed	<0.1

SECTION 4: FIRST AID MEASURES

First aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Inhalation	Move to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Self-Protection of the First Aider	Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	None reasonably foreseeable
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Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically
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SECTION 5: FIREFIGHTING MEASURES

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

Specific Hazards Arising from the Chemical

No information available.

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Explosion Data

Sensitivity to Mechanical Impact None

Sensitivity to Static Discharge None

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Personal Precautions

Ensure adequate ventilation. Use personal protective equipment.

Environmental Precautions

Should not be released into the environment. See Section 12 for additional ecological information.

Methods and Material for Containment and Cleaning Up

Methods for Containment

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Handling

Wear personal protective equipment
Ensure adequate ventilation
Do not get in eyes, on skin, or on clothing
Avoid ingestion and inhalation

Conditions for Safe Storage, Including any Incompatibilities

Storage

Keep container tightly closed in a dry and well-ventilated place
Store at room temperature in the original container
Keep away from direct sunlight

Incompatible Products

No information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS-No	Korea	ACGIH TLV	OSHA PEL
Water	7732-18-5	Not listed	Not listed	Not listed
Ammonium Acetate	631-61-8	Not listed	Not listed	Not listed
Sodium Bisulfite	7631-90-5	TWA: 5 mg/m ³	TWA: 5 mg/m ³	(Vacated) TWA: 5 mg/m ³
Ammonium Oxalate	1113-38-8	Not listed	Not listed	Not listed
Acetone	67-64-1	STEL: 750 ppm TWA: 500 ppm	TWA: 250 ppm STEL: 500 ppm	(Vacated) TWA: 750 ppm (Vacated) TWA: 1800 mg/m ³ (Vacated) STEL: 2400 mg/m ³ (Vacated) STEL: 1000 ppm TWA: 1000 ppm TWA: 2400 mg/m ³
Bathocuproine	52698-84-7	Not listed	Not listed	Not listed

Component	CAS-No	European Union	The United Kingdom	Germany
Water	7732-18-5	Not listed	Not listed	Not listed
Ammonium Acetate	631-61-8	Not listed	Not listed	Not listed
Sodium Bisulfite	7631-90-5	Not listed	STEL: 15 mg/m ³ 15 min TWA: 5 mg/m ³ 8 hr	Not listed
Ammonium Oxalate	1113-38-8	Not listed	Not listed	Not listed

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Acetone	67-64-1	TWA: 500 ppm (8h) TWA: 1210 mg/m ³ (8h)	STEL: 1500 ppm 15 min STEL: 3620 mg/m ³ 15 min TWA: 500 ppm 8 hr TWA: 1210 mg/m ³ 8 hr	TWA: 500 ppm (8 Stunden). AGW - exposure factor 2 TWA: 1200 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 500 ppm (8 Stunden). MAK TWA: 1200 mg/m ³ (8 Stunden). MAK Höhepunkt: 1000 ppm Höhepunkt: 2400 mg/m ³
Bathocuproine	52698-84-7	Not listed	Not listed	Not listed

Biological Exposure Indices

Component	CAS-No	Biological Exposure Indices
Water	7732-18-5	Not listed
Ammonium Acetate	631-61-8	Not listed
Sodium Bisulfite	7631-90-5	Not listed
Ammonium Oxalate	1113-38-8	Not listed
Acetone	67-64-1	25 mg/L Medium: urine Time: end of shift Determinant: Acetone
Bathocuproine	52698-84-7	Not listed

Appropriate engineering controls

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstation location

Individual protection measures, such as personal protective equipment

Personal protective equipment Use only those certified by the Korea Occupational Safety and Health Administration.

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:.
Face-shield.

Skin and Body Protection Wear protective gloves/clothing.

Respiratory Protection None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Colorless to pale brown
Odor	Sharp
Odor Threshold	No information available
pH	7
PH Range	5.5-8.5

Property	Values	Remarks • Method
Melting point/freezing point	No information available	
Boiling Point/Range	106 °C / 222.8 °F	
Flash Point (High in °C)	N/A	
Evaporation Rate	No information available	

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Flammability (solid, gas)	No information available
Flammability Limit in Air	
Upper flammability limit:	No information available
Lower flammability limit:	No information available
Vapor pressure	No information available
Vapor Density	No information available
Specific Gravity	No information available
Water Solubility	Soluble in water
Solubility in other solvents	No information available
Partition coefficient	No information available
Autoignition Temperature	-
Decomposition Temperature	No information available
Kinematic viscosity	No information available
Dynamic viscosity	No information available
Explosive Properties	No information available
Oxidizing Properties	No information available

Other Information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

Reactivity

No Information available

Chemical Stability

Stable under normal conditions

Possibility of Hazardous Reactions

None under normal processing

Conditions to Avoid

Extremes of temperature and direct sunlight.

Incompatible Materials

No information available

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

Product Information

Information on likely routes of exposure

Inhalation	Not an expected route of exposure.
Ingestion	May be harmful if swallowed.
Eyes	Avoid contact with eyes. Irritating to eyes.
Skin	Avoid contact with skin. May cause irritation.

Information on Toxicological Effects

Symptoms	No information available
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Delayed and immediate effects as well as chronic effects from short and long-term exposure

(b) skin corrosion/irritation; Sensitization	Category 2. No information available
Mutagenic Effects	No information available
Carcinogenicity	No information available.
Reproductive Effects	No information available
STOT - single exposure	No information available
STOT - repeated exposure Target Organs	No information available None known.
Aspiration hazard	No information available

Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral)	26194 mg/kg
ATEmix (dermal)	84590 mg/kg
ATEmix (inhalation-dust/mist)	7705.4 mg/L

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity

24.1% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Ammonium Acetate 631-61-8	-	LC50: = 1.06 mg/L, 48h (Cyprinus carpio)	-
Sodium Bisulfite 7631-90-5	-	LC50: = 240 mg/L, 96h static (Gambusia affinis)	EC50: = 119 mg/L, 48h (Daphnia magna)
Acetone 67-64-1	-	LC50: = 8300 mg/L, 96h (Lepomis macrochirus) LC50: 4.74 - 6.33 mL/L, 96h (Oncorhynchus mykiss) LC50: 6210 - 8120 mg/L, 96h static (Pimephales promelas)	EC50: 12600 - 12700 mg/L, 48h (Daphnia magna) EC50: 10294 - 17704 mg/L, 48h Static (Daphnia magna)

Persistence and Degradability

No information available

Bioaccumulation/ Accumulation

No information available

Mobility

No information available.

Component	log Pow
Acetone 67-64-1	-0.24

Other adverse effects

No information available

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SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste Disposal Methods Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport Not Regulated

IATA Not regulated

IMDG/IMO Not regulated
Marine Pollutant Not Applicable

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS-No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	AICS
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	-	X
Ammonium Acetate	631-61-8	KE-01629	X	211-162-9	X	X	-	X	X	X
Sodium Bisulfite	7631-90-5	KE-31484	X	231-548-0	X	X	-	X	X	X
Ammonium Oxalate	1113-38-8	KE-13156	X	214-202-3	X	X	-	X	-	X
Acetone	67-64-1	KE-29367	X	200-662-2	X	X	-	X	X	X
Bathocuproine	52698-84-7	-	X	258-111-7	-	X	-	-	-	-

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	Not applicable	Not applicable	Annex I - Y42
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Water	7732-18-5	Listed	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Listed	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Listed	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Listed	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

Korean National Regulations

Component	CAS-No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration

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Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Annex 1 - KE-01629	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Annex 1 - KE-31484	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Annex 1 - KE-13156	Not applicable	Not applicable
Acetone	67-64-1	Annex 1 - KE-29367	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

Component	CAS-No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Listed	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Listed	Listed	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials

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Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	TWA: 5 mg/m ³	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	STEL: 750 ppm TWA: 500 ppm	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS-No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	Not applicable	Not applicable	2. Group 1 Petroleum (Soluble) 400 L	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS-No	Korea	Biological Exposure Indices
Water	7732-18-5	Not listed	Not listed
Ammonium Acetate	631-61-8	Not listed	Not listed
Sodium Bisulfite	7631-90-5	TWA: 5 mg/m ³	Not listed
Ammonium Oxalate	1113-38-8	Not listed	Not listed
Acetone	67-64-1	STEL: 750 ppm TWA: 500 ppm	25 mg/L Medium: urine Time: end of shift Determinant: Acetone
Bathocuproine	52698-84-7	Not listed	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS-No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Water	7732-18-5	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	Not applicable
Sodium Bisulfite	7631-90-5	Not applicable	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	CAS-No	CERCLA EHS RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Ammonium Acetate	631-61-8	Not applicable	5000 lb	1.0
Sodium Bisulfite	7631-90-5	Not applicable	5000 lb	Not applicable
Ammonium Oxalate	1113-38-8	Not applicable	Not applicable	1.0
Acetone	67-64-1	Not applicable	5000 lb	Not applicable
Bathocuproine	52698-84-7	Not applicable	Not applicable	Not applicable

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CLP Classification - Regulation (EC) No 1272/2008

Warning.

H315 - Causes skin irritation. H319 - Causes serious eye irritation. H335 - May cause respiratory irritation. H336 - May cause drowsiness or dizziness. EUH066 - Repeated exposure may cause skin dryness or cracking.

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water. P337 + P313 - If eye irritation persists: Get medical advice/ attention. P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing. P312 - Call a POISON CENTER or doctor/ physician if you feel unwell. P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection. P332 + P313 - If skin irritation occurs: Get medical advice/ attention.

SECTION 16: OTHER INFORMATION

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.©
Issue Date	05-Sep-2019
Revision Date	05-Sep-2019
Reason for revision	Initial Release.

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

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End of Safety Data Sheet